ST MARY'S DSG, KLOOF

GRADE: 10 DATE: 18 October 2023

PHYSICAL SCIENCE TEST
MAGNETISM, ELECTROSTATICS AND ELECTRICITY

TIME: 40 MINS TOTAL: 44
ASSESSOR: K. Bolton MODERATORS: G. Main and T. Kirsten

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 5 pages. Please check that your question paper is complete.

2. Answer the questions on your double folio sheet provided and number according to the question paper.

3. A data sheet is provided.

FORMULA NOT ON DATA SHEET

$Q=nq_e$

QUESTION 1 – MULTIPLE CHOICE - Write down the letter of the most correct answer.

1.1 Which of the following statements is true concerning magnetic fields?

A They always begin on a North Pole and travel to the South Pole.

B They show both the direction and the strength of the magnetic field.

C They show the direction in which a positive charge will experience force.

D They only cross when multiple magnets are present.

1.2 If a positively charged object is brought near to the disc of a neutral gold leaf electroscope, which of the following will be true?

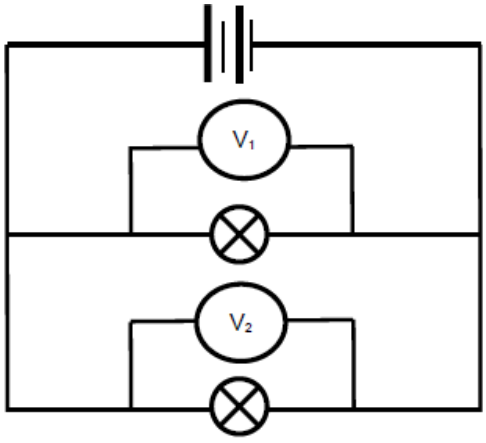
A The gold leaf will collapse against the metal rod.

B The gold leaf will move apart from the metal rod, as they now both have opposite charges to each other.

C The gold leaf will move apart from the metal rod, as they now both have the same charge as each other.

D No change will be observed in the electroscope.

1.3 Two identical light bulbs are connected in parallel, as shown in the circuit diagram below. Voltmeters V_1 and V_2 are connected across each light bulb.



Which ONE of the following voltmeter readings is CORRECT?

A $V_1 = V_2$

B $V_1 = 2V_2$

C $V_1 = ½V_2$

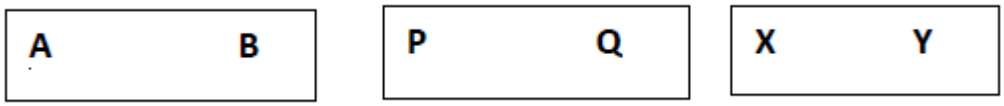
D $V_1 = ¼V_2$

[3x2=6]

QUESTION 2

Study the magnets below, and the information provided:

The field lines inside the first magnet run from A to B
Pole B attracts pole P
Pole Q repels pole X



Using the information provided, answer the following questions:

2.1 List the letters A, P and X and indicate their polarity. (ie North or South) (3)

2.2 Draw the magnetic field pattern observed between pole Q and Pole X. Include all relevant details. (3)

2.3 What would happen to the magnetism of magnet A-B if

(a) You managed to saw the magnet in half? (2)

(b) You repeatedly hammered the magnet? (1)

[9]

QUESTION 3

The earth's magnetic field is shaped like that of a bar magnet around it. The magnetic poles of the earth are not aligned with its geographic poles.

3.1 Comment on this statement: "The North pole is actually a South pole". Is this statement true or false? Justify your answer. (4)

3.2 Migratory birds and sea turtles sense variations in the strength and inclination of the earth's magnetic field as they travel large distances across the world to mating grounds or beaches each year.

One theory which explain how birds manage to sense the magnetic field is that they contain very small particles of iron oxide (magnetite) in their beaks.

Using the domain theory and with reference to the material, describe how the presence of iron oxide in its beak helps a bird to navigate. (3)

[7]

QUESTION 4

Explain each of the following phenomena using your knowledge of electrostatics.

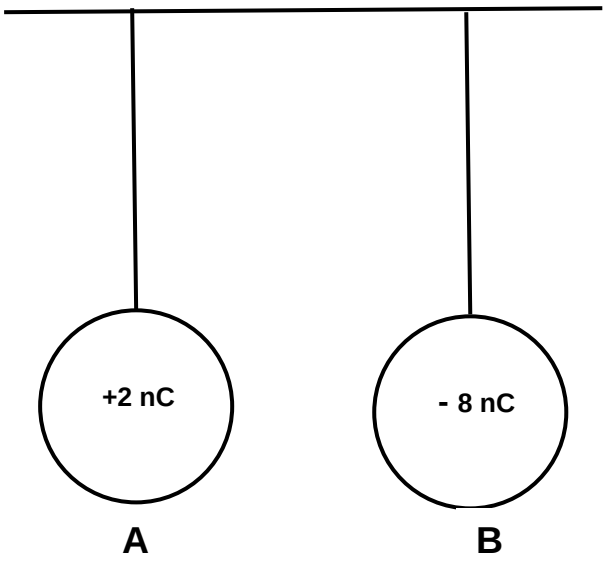
4.1 When the air is very dry, your hand feels a small sharp electric shock when you touch a metal door knob after walking along the carpet to the door. (3)

4.2 When a charged plastic ruler is brought close to small pieces of paper the paper pieces are attracted to the ruler. (3)

[6]

QUESTION 5

Two identical insulated spheres, A and B, suspended by a light inextensible string from a ceiling, are held a distance apart, as shown below.



Sphere A carries a charge of + 2 nC, while sphere B carries a charge of – 8 nC.

Sphere A is brought near the sphere B and the spheres are allowed to touch each other. Immediately after touching, sphere B moves away from sphere A.

5.1 State the *Law of Conservation of charge*. (2)

5.2 Briefly explain why sphere B moves away from sphere A after the spheres have touched. In your answer discuss the movement of charge. (2)

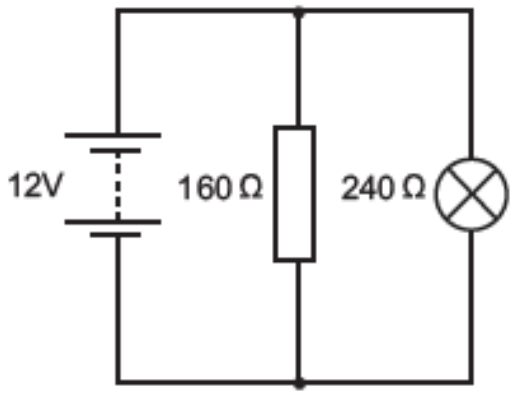
5.3 Calculate the charge on each sphere after they have separated. (2)

5.4 Calculate the number of electrons transferred between the charges. (3)
[9]

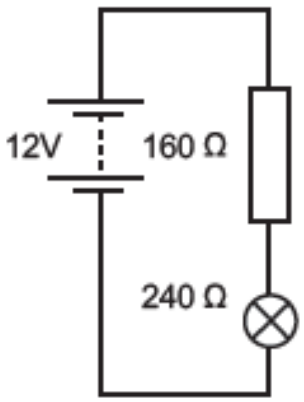
QUESTION 6

Circuit 1 and Circuit 2 consist of a 12 V battery connected to a 160 Ω resistor and a light bulb with a resistance of 240 Ω.

Circuit 1



Circuit 2



6.1 Describe the arrangement of the resistor and light bulb in Circuit 1. (1)

6.2 Calculate the equivalent resistance of Circuit 1. (2)

6.3 Calculate the total current in Circuit 1. (2)

6.4 Calculate the equivalent resistance of Circuit 2. (2)

[7]